MIL-I-3948C
3 September 1982
SUPERSEDING
MIL-I-3948B
14 September 1970

MILITARY SPECIFICATION

INDICATOR TUBE, CARBON MONOXIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of colorimetric indicating tube for the rapid determination of small amounts of carbon monoxide.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat Panel
PPP-B-576	-	Boxes, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard; Corrugated And Solid, Sheet Stock (Container Grade), And Cut Shapes
PPP-T-76	-	Tape, Packaging, Paper (For Carton Sealing)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-D-3945 - Detector Kit; Carbon Monoxide, Colorimetric
MIL-L-10547 - Liners, Case, And Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Munsell Book of Color, Matte Finish

(The Munsell Book of Color may be obtained from the Munsell Color Company, Inc., 2441 N. Calvert Street, Baltimore, MD 21218.)

"Rapid Determination of Small Amounts of Carbon Monoxide," by Martin Shepherd, Analytical Edition of Industrial and Engineering Chemistry, Volume 19, pages 77 to 81, February 1947.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Components.

3.1.1 Tubes. The indicator tubes shall conform to figure 1 or figure 2.

3.1.2 Plugs. To prevent movement and loss of the chemical when the tips are broken, a porous fiber-glass tape shall be used at each end of the tube.

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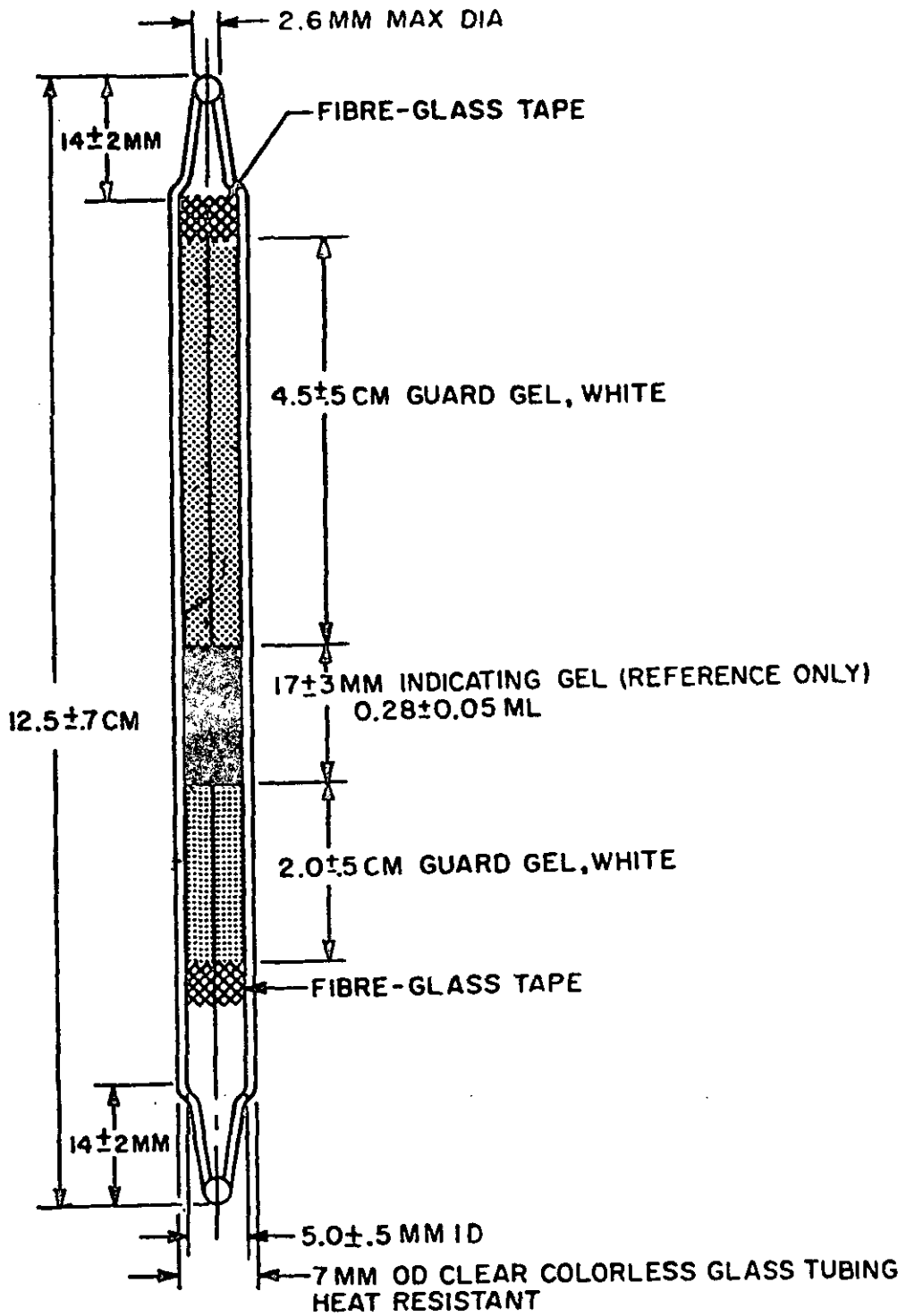


Figure 1. Indicator Tube

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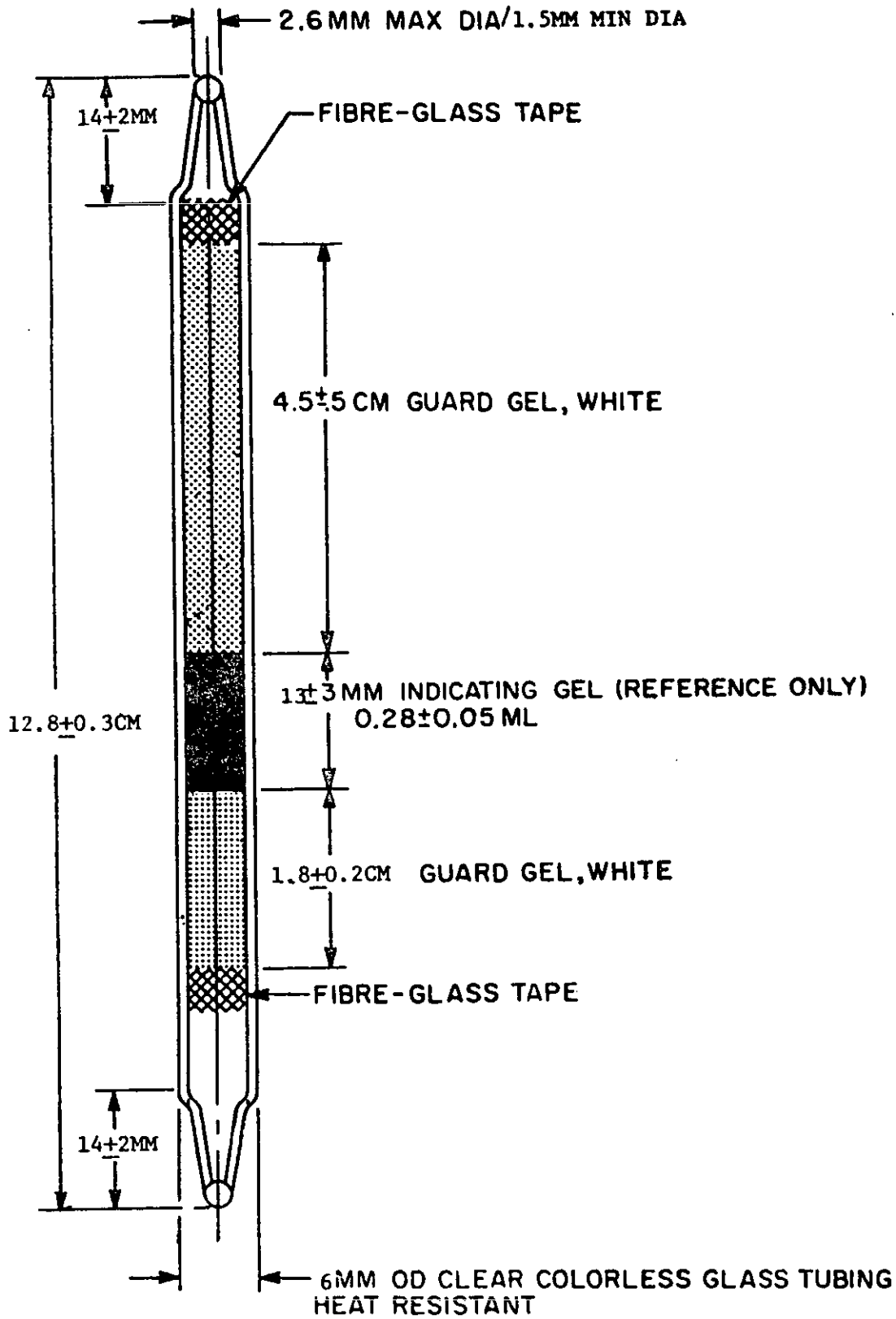


FIGURE 2

Indicator tube

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3.1.3 Reagents. The reagents in the tube shall be in solid form, packed so that mixing of the reagents shall not occur.

3.1.3.1 Indicating gel. The carbon monoxide indicating gel shall be prepared in accordance with directions contained in the article "Rapid Determination of Small Amounts of Carbon Monoxide."

3.1.3.2 Guard chemicals. Sufficient guard chemicals shall be used to protect the carbon monoxide indicating gel from contaminants when tested as specified in 4.2.4.3.

3.1.4 Comparator chart. A color comparator chart for converting the color shade to percent carbon monoxide concentration shall be a part of each unit pack of tubes. Colors shall be stable and of good quality, shall be made of mineral pigments, and shall be in accordance with the Munsell Book of Color. The chart shall be sealed in an envelope of colorless vinyl sheet or laminated between two sheets of clear vinyl to prevent the entrance of moisture during conditions of high humidity. The chart shall conform to figure 3, and shall be of such size as to fit the container specified in section 5.

3.2 Sensitivity.

3.2.1 To 0.0025-percent carbon monoxide. The indicator tube shall detect the presence of 0.0025 percent carbon monoxide when tested as specified in 4.2.4.1.

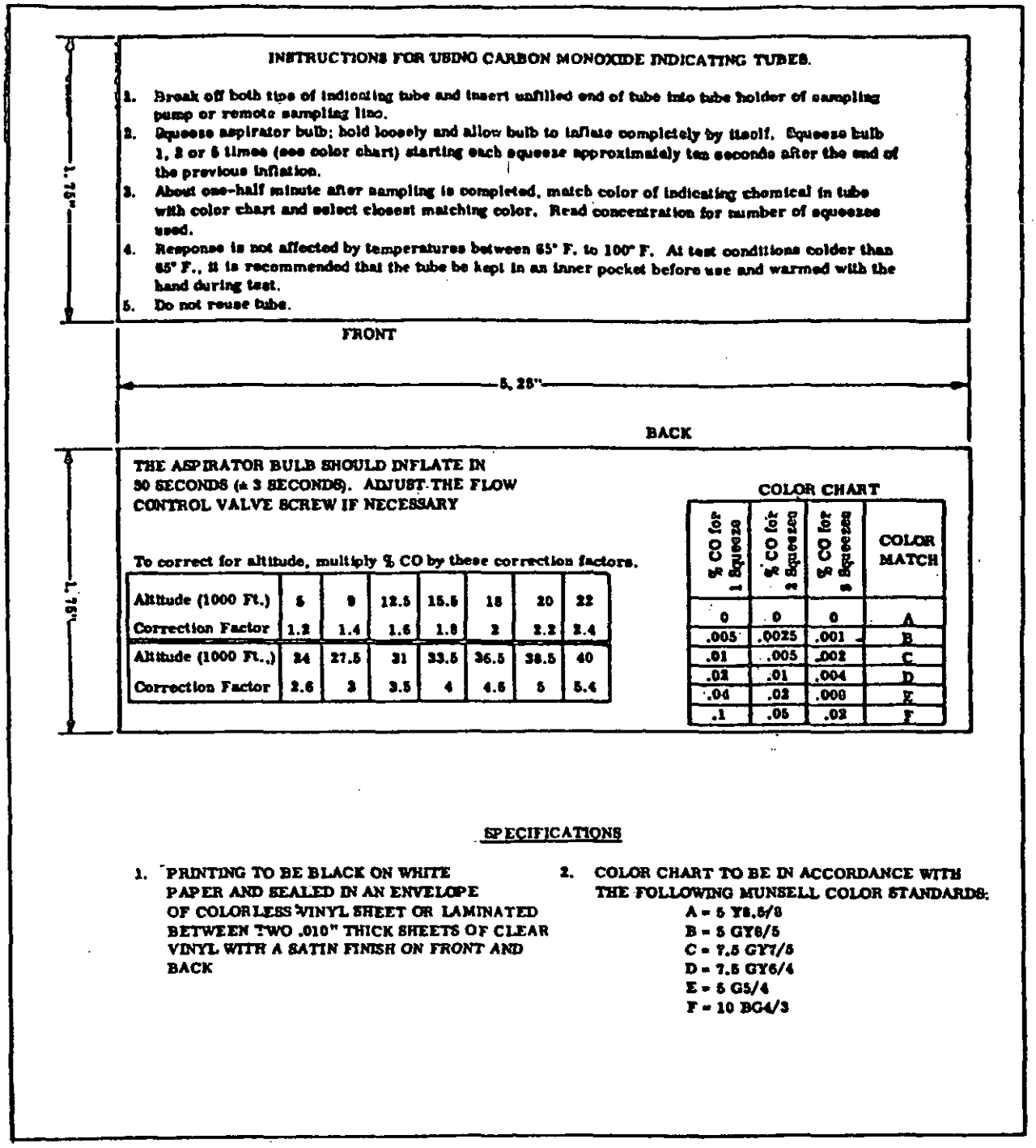
3.2.2 To 0.01-percent carbon monoxide. The indicator tube shall detect the presence of 0.01 percent carbon monoxide when tested as specified in 4.2.4.2.

3.3 Workmanship. The indicating gel shall be free from discoloration. It shall be so packed that there will be no separation between the gel and the glass tape. The tips shall be completely sealed and of such strength as to preclude breakage until they are used for carbon monoxide determination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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SPECIFICATIONS

1. PRINTING TO BE BLACK ON WHITE PAPER AND SEALED IN AN ENVELOPE OF COLORLESS VINYL SHEET OR LAMINATED BETWEEN TWO .010" THICK SHEETS OF CLEAR VINYL WITH A SATIN FINISH ON FRONT AND BACK

2. COLOR CHART TO BE IN ACCORDANCE WITH THE FOLLOWING MUNSELL COLOR STANDARDS:
   - A = 5 YR 5/8
   - B = 5 GY 8/5
   - C = 7.5 GY 7/5
   - D = 7.5 GY 6/4
   - E = 5 G 5/4
   - F = 10 BG 4/3

Figure 3. Comparator Chart

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the carbon monoxide indicator tubes containing guard chemicals and indicating gel, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For examination of tubes and comparator charts. A random sample of indicator tubes and comparator charts shall be selected from each lot in accordance with MIL-STD-105.

4.2.2.3 For tests. A random sample of indicator tubes and comparator charts shall be selected from each lot in accordance with table I.

TABLE I. Sampling for tests

Number of items in lot	Number of sample items to be tested in accordance with		
	4.2.4.1	4.2.4.2	4.2.4.3
6 to 15	2	8	2
16 to 50	3	12	3
51 to 150	5	20	5
151 to 500	8	32	8
501 to 3,200	13	52	13
3,201 to 35,000	20	80	20
35,001 to 500,000	32	128	32
Over 500,000	50	200	50

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing or illegible
(f) Cushioning missing
(g) Comparator chart missing

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(h) Fiberboard pads missing or not as specified
(i) Case liners missing or not as specified (when required)

4.2.3.2 For examination of tubes and comparator charts. The indicator tubes and comparator charts shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Tubes dimensions incorrect (see figure 1 or 2, as applicable)
(b) Reagents missing or not as specified
(c) Fiber-glass tape missing or not as specified
(d) Comparator chart not as specified (as applicable)
(e) Workmanship not as specified (see 3.4)

4.2.3.3 For test. Tubes selected in 4.2.2.3 shall be tested as specified in 4.2.4. Failure of any tube of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. The indicator tubes shall be tested with standard mixtures of 0.01 percent (by volume) carbon monoxide in air at a flow rate of 100 milliliters (ml) per minute, measured by a calibrated flowmeter for a standard exposure of 36 seconds. Each indicator tube, with both tips broken off, shall be placed between a flowmeter and an accurate pressure gage. Air shall pass through the tube at rates up to 120 ml per minute. Back pressure shall not exceed 50.80 millimeters (mm) of mercury. For detection of lower concentrations of carbon monoxide, dilutions of the standard carbon monoxide mixture with air are permissible. Natural daylight shall be used for all readings. Tests shall be conducted as follows:

4.2.4.1 Sensitivity to 0.0025-percent carbon monoxide. A standard mixture of 0.0025 percent carbon monoxide in air shall be passed through the sample indicator tubes at a rate of 100 ml per minute for 36 seconds at sea level. The gas and tube shall be at 24° ± 3°C. There shall be a perceptible color change in the indicator gel.

4.2.4.2 Sensitivity to 0.01-percent carbon monoxide. Divide the number of sample indicator tubes to be tested in fourths. A standard mixture of 0.01 percent carbon monoxide in air shall be passed through the sample indicator tubes at a rate of 100 ml per minute for 18, 36, 72, and 144 seconds for the first, second, third, and last fourth respectively. The gas and tube shall be at 24° ± 3°C. The resulting indications given by the tubes, shall match the figure 3 color standards B, C, D, and E, respectively.

4.2.4.3 Guard chemicals. Air saturated with water vapor at 25°C and 760 mm mercury pressure and containing 0.02 percent aviation gasoline vapor by volume shall be drawn through the indicator tube at a sampling rate of 5 ml per minute for a period of 1 hour. The indicator tube shall then pass the test specified in 4.2.4.2 using the 72-second procedure.

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5. PACKAGING

5.1 Unit packing. Unit packing shall be level A or industrial as specified (see 6.2).

5.1.1 Level A. Unit packing shall meet the requirements of MIL-P-116, method III. Twelve tubes shall be cushioned and unit packed within a partitioned pencil-type box not more than 5-1/2 inches long, 2.0 inches wide, and 13/16 inch high. Each box shall contain a comparator chart as specified in 3.1.4.

5.1.1.1 Intermediate packing. Like quantities of level A unit packs (see 5.1.1) for the intermediate pack shall be at the option of the contractor. Weight limitations of the contractor shall be adhered to. The intermediate package shall be a close-fitting fiberboard box conforming to PPP-B-636, class domestic. Motion of contents within the box shall be prevented by inserting fiberboard pads as needed. The fiberboard shall conform to PPP-F-320, class domestic. The box shall be closed and sealed in accordance with the requirements specified in the appendix of PPP-B-636 for interior boxes.

5.1.2 Industrial. Tubes shall be unit packed commercially in accordance with MIL-STD-1188.

5.2 Packing. Tubes shall be packed level A, B, or industrial as specified (see 6.2).

5.2.1 Level A. Like quantities of tubes intermediately packed as specified in 5.1.1.1 shall be packed in close-fitting, overseas type, exterior containers conforming to PPP-B-576, PPP-B-585, or PPP-B-601. Plywood when used, shall conform to NN-P-530. Motion of contents shall be prevented by inserting pads of fiberboard where needed. Fiberboard shall conform to PPP-F-320, grade V2s or V3c. When the containers being overpacked within the exterior containers are not water-resistant, case liners shall be provided which are fabricated in accordance with MIL-L-10547 and the appendix thereto. Boxes shall be closed and strapped in accordance with the appendix of the applicable specification.

5.2.2 Level B. Like quantities of tubes intermediately packed as specified in 5.1.1.1 shall be packed in domestic type exterior containers conforming to PPP-B-576, PPP-B-585, PPP-B-591, PPP-B-601, or grade V3c boxes of PPP-B-636. Wood boxes shall be closed and strapped in accordance with the closure requirements specified in the appendix of the applicable specification. The fiberboard box shall be closed in accordance with the closure requirements specified in PPP-B-636. In addition, all seams including the manufacturers joint shall be sealed with tape conforming to PPP-T-76.

5.2.3 Industrial. Tubes shall be commercially packed in accordance with MIL-STD-1188.

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5.3 Unitization. When specified (see 6.2), packs shall be palletized in accordance with MIL-STD-147.

5.4 Marking. Level A and B unit packs, intermediate packs, and packs shall be marked in accordance with MIL-STD-129. Commercial unit packs, intermediate packs and packs shall be marked in accordance with MIL-STD-1188. In addition the marking shall also include the following:

(a) Date of manufacture,
(b) Expiration date: 2 years from the month and year of tube sealing.

6. NOTES

6.1 Intended use. The colorimetric indicator tubes covered by this specification are intended for the rapid colorimetric determination of small or physiologically significant amounts of carbon monoxide. Bureau of Medicine and Surgery, Department of the Navy, does not consider tubes covered by this specification acceptable for Navy use in monitoring carbon monoxide levels in conjunction with any type of health hazard assessment.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of unit packing and packing required (see 5.1 and 5.2).
(c) If palletization is required (see 5.3).

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodian:

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